Homework 4 Solutions

Based on Lecture 4: Inverse and Logarithmic Functions.

Exercise 1 (one-to-one). Use the horizontal line test or an algebraic reason to decide whether each function is one-to-one on the stated domain.

- $f(x) = 4x - 1$ on $\mathbb{R}$
- $g(x) = x^2 - 3$ on $\mathbb{R}$
- $h(x) = x^2 - 3$ on $[0, \infty)$

$f(x) = 4x - 1$ on $\mathbb{R}$. Suppose $f(x_1) = f(x_2)$. Then $4x_1 - 1 = 4x_2 - 1$, so $4x_1 = 4x_2$, so $x_1 = x_2$. Equal outputs force equal inputs, so f is $\boxed{\text{one-to-one}}$. (Graphically, every horizontal line $y = c$ meets the line $4x - 1 = c$ at the single point $x = (c + 1)/4$.)

$g(x) = x^2 - 3$ on $\mathbb{R}$. $g(2) = 4 - 3 = 1$ and $g(-2) = 4 - 3 = 1$, so the two distinct inputs 2 and -2 give the same output. Thus g is $\boxed{\text{not one-to-one}}$. (The horizontal line $y = 1$ crosses the graph twice, at $x = 2$ and $x = -2$.)

$h(x) = x^2 - 3$ on $[0, \infty)$. Suppose $x_1, x_2 \geq 0$ and $h(x_1) = h(x_2)$, i.e. $x_1^2 - 3 = x_2^2 - 3$, so $x_1^2 = x_2^2$, so $x_1 = \pm x_2$. Since $x_1, x_2 \geq 0$, the only possibility is $x_1 = x_2$ (if $x_1 = -x_2$ with both nonnegative, then $x_1 = x_2 = 0$, which is a special case of $x_1 = x_2$ anyway). Thus h is $\boxed{\text{one-to-one}}$ on this restricted domain, even though the same formula failed to be one-to-one on all of $\mathbb{R}$.

Exercise 2 (inverse). Find the inverse of $f(x) = \dfrac{x + 4}{3}$, then verify both compositions.

Write $y = \dfrac{x + 4}{3}$ and solve for x:

$$3y = x + 4 \quad \Longrightarrow \quad x = 3y - 4.$$

Swap the variable names. Thus

$$\boxed{f^{-1}(x) = 3x - 4}.$$

Check both compositions:

$$f^{-1}(f(x)) = 3\left(\frac{x + 4}{3}\right) - 4 = (x + 4) - 4 = x,$$

$$f(f^{-1}(x)) = \frac{(3x - 4) + 4}{3} = \frac{3x}{3} = x.$$

Both equal x, confirming the inverse.

Exercise 3 (restricted inverse). Restrict $f(x) = (x - 1)^2$ to $x \geq 1$. Find $f^{-1}(x)$ and state the domain and range of both f and f^{-1}.

Domain and range of f. By restriction, $\text{dom}(f) = [1, \infty)$. For $x \geq 1$, $x - 1 \geq 0$, so $(x - 1)^2$ ranges over every value in $[0, \infty)$ as $x - 1$ ranges over $[0, \infty)$; hence $\text{range}(f) = [0, \infty)$.

Finding the inverse. Write $y = (x - 1)^2$ with $x \geq 1$, and solve for x. Taking square roots of both sides gives $\sqrt{y} = |x - 1|$. Since $x \geq 1$ means $x - 1 \geq 0$, we have $|x - 1| = x - 1$, so

$$\sqrt{y} = x - 1 \quad \Longrightarrow \quad x = 1 + \sqrt{y}.$$

Swap the variable names. Thus

$$\boxed{f^{-1}(x) = 1 + \sqrt{x}}.$$

Domain and range of f^{-1}. Since $\text{dom}(f^{-1}) = \text{range}(f)$ and $\text{range}(f^{-1}) = \text{dom}(f)$, we get $\text{dom}(f^{-1}) = [0, \infty)$ and $\text{range}(f^{-1}) = [1, \infty)$. (Directly: for $x \geq 0$, $1 + \sqrt{x} \geq 1$, matching $\text{dom}(f) = [1, \infty)$.)

Check: $f^{-1}(f(x)) = 1 + \sqrt{(x - 1)^2} = 1 + |x - 1| = 1 + (x - 1) = x$ for $x \geq 1$, and $f(f^{-1}(x)) = \left((1 + \sqrt{x}) - 1\right)^2 = (\sqrt{x})^2 = x$ for $x \geq 0$.

Exercise 4 (translation). Rewrite each statement in the other form.

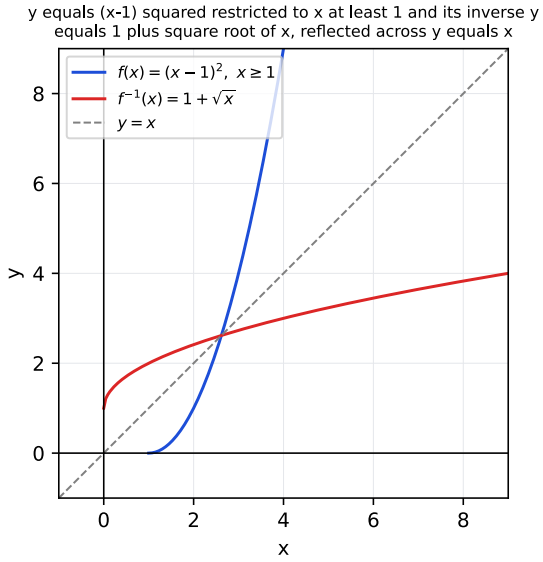


Figure 1: The graph of y equals (x-1) squared restricted to x at least 1 and its inverse y equals 1 plus square root of x, reflected across y equals x

- $4^3 = 64$
- $10^{-4} = 0.0001$
- $\log_7 49 = 2$
- $\ln(e^{-3}) = -3$

Using $\log_b x = y \iff b^y = x$:

- $\boxed{\log_4 64 = 3}$
- $\boxed{\log_{10}(0.0001) = -4}$
- $\boxed{7^2 = 49}$
- $\boxed{e^{-3} = e^{-3}}$ (the exponential form of a natural-log statement about e^{-3} is simply the tautology $e^{-3} = e^{-3}$, since the base of ln is e and the stated exponent is already -3; this confirms the original statement is consistent with the definition).

Exercise 5 (exact values). Evaluate without a calculator.

- $\log_2 32$
- $\log_9 3$
- $\log_{10}(0.01)$
- $\ln 1$

Each value is the exponent that produces the argument:

- $2^5 = 32$, so $\boxed{\log_2 32 = 5}$.
- $9^{1/2} = \sqrt{9} = 3$, so $\boxed{\log_9 3 = \frac{1}{2}}$.
- $10^{-2} = 0.01$, so $\boxed{\log_{10}(0.01) = -2}$.
- $e^0 = 1$, so $\boxed{\ln 1 = 0}$.

Exercise 6 (features). State the domain, range, x-intercept, vertical asymptote, and monotonicity of $y = \log_{1/3} x$, then plot the graph, labeling the asymptote.

Here $b = \frac{1}{3}$, which satisfies $0 < b < 1$:

- Domain: $\boxed{(0, \infty)}$
- Range: $\boxed{(-\infty, \infty)}$
- x-intercept: $\boxed{(1, 0)}$ (since $\log_b 1 = 0$ for any valid base b)
- Vertical asymptote: $\boxed{x = 0}$
- Monotonicity: since $0 < b < 1$, $y = \log_{1/3} x$ is $\boxed{\text{decreasing}}$

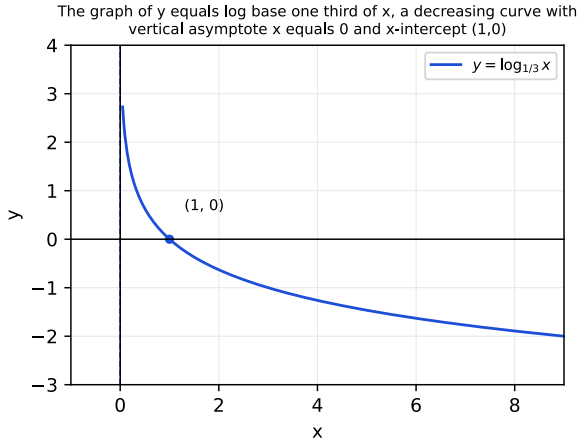


Figure 2: The graph of y equals log base one third of x, a decreasing curve with vertical asymptote x equals 0 and x-intercept (1,0)

Exercise 7 (log rules). Expand each expression as far as possible. Assume all variables are positive.

- $\log_5 \left(\dfrac{25x}{z^2} \right)$
- $\ln \left(\dfrac{\sqrt{x}}{3y} \right)$
- $\log_2(16a^4b)$

(a) Using the quotient rule, then the product rule, then the power rule:

$$\log_5 \left(\frac{25x}{z^2} \right) = \log_5(25x) - \log_5(z^2) = \log_5 25 + \log_5 x - 2\log_5 z = \boxed{2 + \log_5 x - 2\log_5 z}.$$

(b) Writing $\sqrt{x} = x^{1/2}$ and using the quotient, product, and power rules:

$$\ln \left(\frac{\sqrt{x}}{3y} \right) = \ln \sqrt{x} - \ln(3y) = \tfrac{1}{2} \ln x - (\ln 3 + \ln y) = \boxed{\tfrac{1}{2} \ln x - \ln 3 - \ln y}.$$

(c) Using the product rule, then the power rule, and $16 = 2^4$:

$$\log_2(16a^4b) = \log_2 16 + \log_2 a^4 + \log_2 b = 4 + 4\log_2 a + \log_2 b = \boxed{4 + 4\log_2 a + \log_2 b}.$$

Exercise 8 (exact equations). Solve exactly.

- $3^{2x-1} = 27$
- $\log_2(x - 1) = 4$

- $\ln(x+2) = 0$

(a) Write $27 = 3^3$, so both sides share base 3:

$$3^{2x-1} = 3^3 \quad \Longrightarrow \quad 2x - 1 = 3 \quad \Longrightarrow \quad 2x = 4 \quad \Longrightarrow \quad \boxed{x = 2}.$$

(b) Convert to exponential form:

$$\log_2(x-1) = 4 \quad \Longrightarrow \quad x - 1 = 2^4 = 16 \quad \Longrightarrow \quad \boxed{x = 17}.$$

(Domain check: $x - 1 = 16 > 0$. Valid.)

(c) Convert to exponential form:

$$\ln(x+2) = 0 \quad \Longrightarrow \quad x + 2 = e^0 = 1 \quad \Longrightarrow \quad \boxed{x = -1}.$$

(Domain check: $x + 2 = 1 > 0$. Valid.)

Exercise 9 (extraneous solutions). Solve each equation, then check every candidate solution against the domain of the original logarithms and discard any extraneous root.

- $\log_3(x+2) + \log_3 x = 1$
- $\log(x) + \log(x-3) = 1$

(a) The original equation requires $x + 2 > 0$ and $x > 0$, i.e. $x > 0$. Combine with the product rule, then convert to exponential form:

$$\log_3\left[x(x+2)\right] = 1 \quad \Longrightarrow \quad x^2 + 2x = 3^1 = 3 \quad \Longrightarrow \quad x^2 + 2x - 3 = 0 \quad \Longrightarrow \quad (x+3)(x-1) = 0,$$

so $x = -3$ or $x = 1$. Since $x = -3$ fails $x > 0$, it is extraneous and rejected. Only $\boxed{x = 1}$ solves the original equation. (Check: $\log_3 3 + \log_3 1 = 1 + 0 = 1$.)

(b) (Base 10.) The original equation requires $x > 0$ and $x - 3 > 0$, i.e. $x > 3$. Combine with the product rule, then convert to exponential form:

$$\log\left[x(x-3)\right] = 1 \quad \Longrightarrow \quad x^2 - 3x = 10^1 = 10 \quad \Longrightarrow \quad x^2 - 3x - 10 = 0 \quad \Longrightarrow \quad (x-5)(x+2) = 0,$$

so $x = 5$ or $x = -2$. Since $x = -2$ fails $x > 3$, it is extraneous and rejected. Only $\boxed{x = 5}$ solves the original equation. (Check: $\log 5 + \log 2 = \log 10 = 1$.)

Exercise 10 (calculator equations). Use a calculator and round each answer to three decimal places.

(a) $2^x = 7$**.** Take ln of both sides:

$$x \ln 2 = \ln 7 \quad \Longrightarrow \quad x = \frac{\ln 7}{\ln 2} = \frac{1.945910}{0.693147} \approx \boxed{2.807}.$$

(b) $5e^{0.4t} = 60$**.** Divide, then take ln:

$$e^{0.4t} = 12 \quad \Longrightarrow \quad 0.4t = \ln 12 \quad \Longrightarrow \quad t = \frac{\ln 12}{0.4} = \frac{2.484907}{0.4} \approx \boxed{6.212}.$$

(c) $\log_3(x+1) = 2.7$**.** Convert to exponential form:

$$x + 1 = 3^{2.7} = e^{2.7 \ln 3} = e^{2.966253} \approx 19.41920 \quad \Longrightarrow \quad x \approx \boxed{18.419}.$$

Exercise 11 (features). For $y = -2\log_3(x+1) + 4$, describe the transformations from $y = \log_3 x$, then state the domain, vertical asymptote, and the point corresponding to $(1, 0)$.

Transformations. Starting from $y = \log_3 x$: replace x by $x + 1$ to shift left 1; multiply by 2 to scale vertically; the minus sign reflects the result across the x-axis; finally add 4 to shift up 4.

Domain and asymptote. The argument requires $x + 1 > 0$, i.e. $x > -1$, so the domain is $\boxed{(-1, \infty)}$ and the vertical asymptote is $\boxed{x = -1}$ (the vertical shift by 4 does not move the asymptote).

Image of $(1, 0)$**.** A point (a, b) on $y = \log_3 x$ maps, under $x \mapsto x - 1$ (i.e. a *left* shift by 1, so $h = -1$ in the transformation $x - h$) and $y \mapsto -2y + 4$, to $(a + h, -2b + 4) = (a - 1, -2b + 4)$. With $(a, b) = (1, 0)$, the image is $(1 - 1, -2 \cdot 0 + 4) = \boxed{(0, 4)}$. (Check directly: at $x = 0$, $y = -2\log_3(0 + 1) + 4 = -2 \cdot 0 + 4 = 4$, matching.)

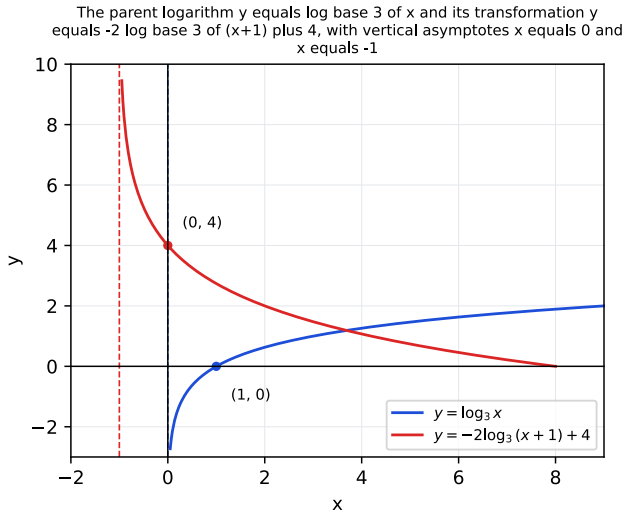


Figure 3: The parent logarithm y equals log base 3 of x and its transformation y equals -2 log base 3 of (x+1) plus 4, with vertical asymptotes x equals 0 and x equals -1

Exercise 12 (sketch). Sketch $y = \ln(x - 3)$, labeling its vertical asymptote and two points obtained from $y = \ln x$.

Starting from $y = \ln x$, shift right 3. The vertical asymptote moves from $x = 0$ to $\boxed{x = 3}$, and the domain becomes $x > 3$. Two convenient points on $y = \ln x$ are $(1, 0)$ and $(e, 1)$ (since $\ln 1 = 0$ and $\ln e = 1$); shifting each right 3 gives $(4, 0)$ and $(e + 3, 1) \approx (5.718, 1)$ on $y = \ln(x - 3)$.

Exercise 13 (challenge). Explain why $y = \log_2(x^2)$ is not the same function as $y = 2\log_2 x$. State the domain of each function and identify where their formulas agree.

Domains. For $y = \log_2(x^2)$, the argument x^2 is positive for every $x \neq 0$, so the domain is $\boxed{(-\infty, 0) \cup (0, \infty)}$. For $y = 2\log_2 x$, the argument x itself must be positive, so the domain is $\boxed{(0, \infty)}$.

Why the functions differ. Two functions are equal only if they share the same domain and agree on it. Here $\log_2(x^2)$ is defined at, say, $x = -1$ (giving $\log_2 1 = 0$), while $2\log_2 x$ is undefined there (since $\log_2(-1)$ does not exist). Because one function has points in its domain where the other is undefined, they are **not the same function**, even though their formulas look related.

Where the formulas agree. By the power rule applied to $|x|$, $\log_2(x^2) = \log_2(|x|^2) = 2\log_2|x|$ for every $x \neq 0$. For $x > 0$, $|x| = x$, so $2\log_2|x| = 2\log_2 x$ — the two formulas coincide. For $x < 0$, $2\log_2|x| = 2\log_2(-x)$ is a real number, but $2\log_2 x$ is undefined there. Hence the formulas agree exactly on $\boxed{(0, \infty)}$, the (smaller) domain of $y = 2\log_2 x$.

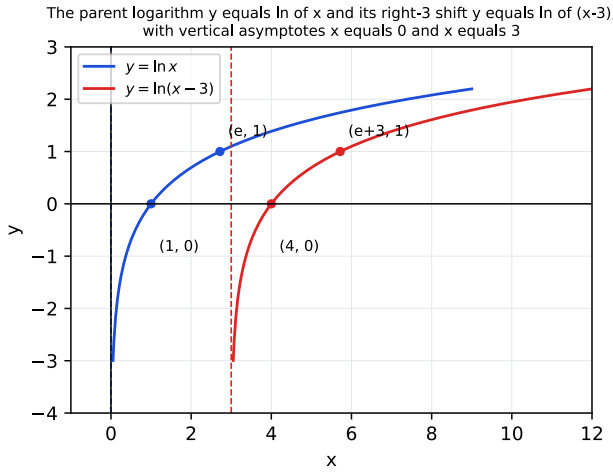


Figure 4: The parent logarithm y equals ln of x and its right-3 shift y equals ln of (x-3), with vertical asymptotes x equals 0 and x equals 3

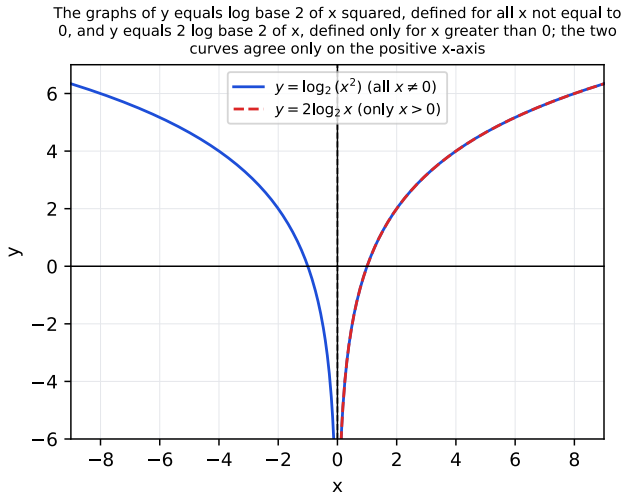


Figure 5: The graphs of y equals log base 2 of x squared, defined for all x not equal to 0, and y equals 2 log base 2 of x, defined only for x greater than 0; the two curves agree only on the positive x-axis

Exercise 14 (pH). A solution has $[\mathrm{H}^+] = 10^{-6.2}$. Find its pH. How many times greater is its hydrogen-ion concentration than a solution with pH 8.2?

pH of the first solution.

$$\mathrm{pH} = -\log_{10}(10^{-6.2}) = -(-6.2) = \boxed{6.2}.$$

Comparing concentrations. A solution with pH 8.2 has $[\mathrm{H}^+]_2 = 10^{-8.2}$ (from $\mathrm{pH} = -\log_{10}[\mathrm{H}^+]$ solved for the concentration: $[\mathrm{H}^+] = 10^{-\mathrm{pH}}$). The ratio of concentrations is

$$\frac{[\mathrm{H}^+]_1}{[\mathrm{H}^+]_2} = \frac{10^{-6.2}}{10^{-8.2}} = 10^{(-6.2)-(-8.2)} = 10^2 = \boxed{100}.$$

The pH-6.2 solution has 100 times the hydrogen-ion concentration of the pH-8.2 solution.

Exercise 15 (decibels). A sound changes from 50 dB to 80 dB. By what factor does its intensity change? Explain using the formula for L.

From $L = 10\log_{10}(I/I_0)$, solve for the intensity ratio: $\log_{10}(I/I_0) = L/10$, so $I/I_0 = 10^{L/10}$. Thus

$$\frac{I_1}{I_0} = 10^{50/10} = 10^5, \qquad \frac{I_2}{I_0} = 10^{80/10} = 10^8.$$

The factor by which intensity changes is

$$\frac{I_2}{I_1} = \frac{I_2/I_0}{I_1/I_0} = \frac{10^8}{10^5} = 10^3 = \boxed{1000}.$$

A 30-decibel increase is three 10-decibel steps, and each 10-decibel step multiplies intensity by 10 (since $10\log_{10} 10 = 10$), so 30 decibels multiplies intensity by $10^3 = 1000$.

Exercise 16 (calculator application). A \$5000 investment earns 5% annual interest compounded continuously. Use logarithms to find how long it takes to reach \$8000. Round to two decimal places.

Using $t = \dfrac{\ln(A/P)}{r}$ with $A = 8000$, $P = 5000$, $r = 0.05$:

$$t = \frac{\ln(8000/5000)}{0.05} = \frac{\ln(1.6)}{0.05} = \frac{0.470004}{0.05} \approx \boxed{9.40 \text{ years}}.$$

Exercise 17 (modeling, calculator). A bacteria culture follows $P(t) = 300 \cdot 2^{t/4}$, where t is measured in hours. Use logarithms to find when the population first reaches 5000. Give an exact logarithmic expression and a calculator approximation in hours.

Set $P(t) = 5000$ and solve for t:

$$300 \cdot 2^{t/4} = 5000 \quad \Longrightarrow \quad 2^{t/4} = \frac{5000}{300} = \frac{50}{3}.$$

Take $\log_2$ of both sides:

$$\frac{t}{4} = \log_2\!\left(\frac{50}{3}\right) \quad \Longrightarrow \quad \boxed{t = 4\log_2\!\left(\frac{50}{3}\right)}.$$

Using the change-of-base formula $\log_2\!\left(\frac{50}{3}\right) = \dfrac{\ln(50/3)}{\ln 2} = \dfrac{2.813411}{0.693147} \approx 4.058894$:

$$t \approx 4(4.058894) \approx \boxed{16.24 \text{ hours}}.$$

Exercise 18 (challenge). Let $f(x) = a^x$ for $a > 0$, $a \neq 1$. Prove that $f^{-1}(x) = \log_a x$ directly by checking both compositions, including the necessary domain restrictions.

Step 1: f is one-to-one, so an inverse exists. Suppose $a > 1$ and $x_1 < x_2$. Exponential functions with base greater than 1 are increasing, so $a^{x_1} < a^{x_2}$, i.e. $f(x_1) \neq f(x_2)$. Suppose instead $0 < a < 1$ and $x_1 < x_2$; then a^x is decreasing, so $a^{x_1} > a^{x_2}$, again $f(x_1) \neq f(x_2)$. In either case distinct inputs give distinct outputs, so f is one-to-one on $\mathbb{R}$ and an inverse function exists. Also $\text{dom}(f) = \mathbb{R}$ and, since $a^x > 0$ for every real x and every positive target value x_0 is achieved by $f(\log_a x_0)$ (verified in Step 2 below), $\text{range}(f) = (0, \infty)$.

Step 2: define $g(x) = \log_a x$ on $x > 0$ and check $f(g(x)) = x$. By the definition of the logarithm, $\log_a x = y$ means exactly $a^y = x$. Setting $y = \log_a x$, this definition directly gives $a^{\log_a x} = x$, i.e.

$$f(g(x)) = a^{\log_a x} = x \qquad \text{for all } x > 0 = \text{dom}(g).$$

Step 3: check $g(f(x)) = x$. We must show $\log_a(a^x) = x$ for every real x. By definition, $\log_a(a^x)$ is the unique real number t satisfying $a^t = a^x$. Since $f(t) = a^t$ is one-to-one (Step 1), the equation $a^t = a^x$ has exactly one solution, namely $t = x$. Hence

$$g(f(x)) = \log_a(a^x) = x \qquad \text{for all } x \in \mathbb{R} = \text{dom}(f).$$

Conclusion. Both compositions return the identity on the correct domains: $f \circ g = \text{id}$ on $\text{dom}(g) = (0, \infty) = \text{range}(f)$, and $g \circ f = \text{id}$ on $\text{dom}(f) = \mathbb{R} = \text{range}(g)$. This is exactly the defining property of an inverse function, so

$$\boxed{f^{-1}(x) = \log_a x, \qquad \text{dom}(f^{-1}) = (0, \infty), \qquad \text{range}(f^{-1}) = \mathbb{R}.}$$